VI) Patterns and Modeling (Try 2)

Utilize the figures below for problems 9 - 12. You may assume they are regular decagons with side lengths of 1 unit each.

Figure 1

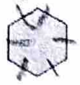


Figure 2

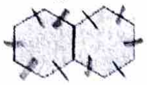
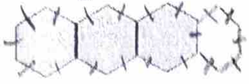


Figure 3



9) Complete the table below: (1 pt)

Figure	1	2	3
Perimeter	6	10	14

10) Is there a common difference? If so, identify its value. If not, explain why. (1 pt)

-½

yes and the common difference is 4 ~~and it starts from 6~~

11) Let 'n' represent the figure number. Write an expression using 'n' that could be utilized to determine the perimeter of the nth figure. (1 pt)

-½

$(6+4n)-4$

12) Use your expression from item 11 to determine the perimeter of the 30th figure. (1 pt)

$(6+(4\cdot30)$

$(6+120)-4$

$126-4$

122

Perimeter of 30th figure

122 unit

13) Model and Solve using "bags" ~~and~~ "cubes":

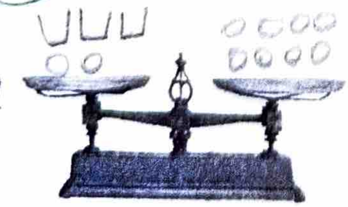
-½

$3x + 2 = 8$ (1 pt)

$-2 \quad -2$

$\frac{3x}{3} = \frac{6}{3}$

$\boxed{x=2}$



Overall Score For This Standard: 3½

VII) Solve Linear Equations (Try 1)

For problems 14 - 17, solve for x. Reminder: Show work!

14) $2(x + 5) = 2x + 10$ (1 pt)

$2x+10 = 2x+10$

$-2x \quad -2x$

$10=10$

Infinite Solutions

15) $12x + 11 = 2x + 11 + x$ (1 pt)

-1

$12x+11 = 3x+11$

$-3x \quad -3x$

$9x+11 = 11$

$-11 \quad -11$

$\frac{9x}{9} = \frac{0}{9}$

$x=0$

no ~~solutions~~

16) $1 + 2(x + 1) = x + 5$ (1 ½ pts)

$1+2x+2 = x+5$

$2x+3 = x+5$

$-x \quad -x$

$1x+3 = 5$

$-3 \quad -3$

$x= 2$

One Solution

17) $4x + 24 = 5(x + 5) + 3x$ (1 ½ pts)

-½

$4x+24 = 5x+5+3x$

$4x+24 = 8x+5$

$-5 \quad -5$

$4x+19 = 8x$

$-4x \quad -4x$

$\frac{19}{4} = \frac{4x}{4}$

$4\overline{)19.00}$ = 4.75; 16, 30, 28, 20

$x= 4.75$

One Solution

Overall Score For This Standard: 3½

Math 3
CC 8: the OCHO

Name: Arjun Rakesh
Date:______________ Per:____

Directions: Show work for all problems involving calculations. All answers must be in simplest form.

IV) Exponent Properties (Try 4)

For problems 1 - 4, simplify the expression. Ensure your final answer has only positive exponents.

1) $(x^6)(x^4)$ (½ pt)

$x^6 \cdot x^4$

$x^{6+4} \rightarrow x^{10}$

2) $\frac{x^{40}}{x^5}$ (½ pt)

$x^{40} - x^5$

x^{35}

3) $(x^7)^{10} \cdot x^{14}$ (1 ½ pts)

$(x^{7 \cdot 10}) \cdot x^{14}$

$x^{70} \cdot x^{14}$

x^{84}

4) $\left(\frac{x^9y^{-15}}{x^{-7}y^{11}}\right)$ (1 ½ pts)

$\frac{x^9}{x^{-7}} \cdot \frac{y^{-15}}{y^{11}}$

$(x^{9+7}) \cdot (y^{-15-11}) \rightarrow x^{16} \cdot y^{-25}$

$\boxed{\frac{x^{16}}{y^{25}}} \leftarrow x^{16} \cdot \frac{1}{y^{25}}$

5) Solve for x: (1 pt)

$(4^2)^x = 4^{33}$

$4^{2x} = 4^{33}$

$\frac{2x}{2} = \frac{33}{2}$

$x = 16.5$

$2 \overline{)33.0}$ = 16.5
−2
13
−12
10

Overall Score For This Standard: 5

V) Scientific Notation (Try 3)

6) Convert 60,000,000 to scientific notation (1 pt)

60,000,000

6×10^7

7) Simplify the expression, write your final answer in scientific notation. (2 pts)

$-\frac{1}{2}$

$(6.2 \times 10^{15}) \cdot (2 \times 10^3)$

$(6.2 \cdot 2) \cdot 10^{15+3}$

$12.4 \cdot 10^{18}$

1.24×10^{19}

8) Last year, Bobissa was able to eat all her Halloween candy in 8.4×10^4 seconds. This year, she did it in 4×10^2 seconds. How much longer did it take her to finish last year compared to this year? Ensure your your answer is in scientific notation. (2 pts)

$(8.4 \times 10^4) - (4 \times 10^2)$

$(8.4 - 0.04) \times 10^4$

$\boxed{8.36 \times 10^4}$

8.40
−0.04
8.36

It took her 8.36×10^4 more seconds last year then this year to eat her candies

Overall Score For Standard: 4½